PKCERT AI & Software Development Internship

Task 15 - Model Persistence & Mini-Project

Objective: Save and load machine learning models with Pickle and Joblib, then complete an end-to-end machine learning pipeline with EDA, preprocessing, model training, evaluation, and final model persistence.

Part A - Saving Trained Models

A binary classification model was trained on the Breast Cancer Wisconsin Diagnostic dataset. The model uses a StandardScaler followed by LogisticRegression, which is appropriate because the dataset contains continuous numeric measurements with different ranges.

Item	Details
Dataset	Breast Cancer Wisconsin Diagnostic
Problem Type	Binary classification
Model	StandardScaler + LogisticRegression
Pickle File	outputs/models/part_a_breast_cancer_logreg.pkl
Joblib File	outputs/models/part_a_breast_cancer_logreg.joblib

The same trained model object was saved twice: once with Python's pickle module and once with joblib. This satisfies the requirement to persist a trained model in both formats.

Part B - Loading & Verifying Models

Both saved files were loaded back into memory and tested on the same held-out test set used for the original model. The predictions from the loaded models matched the original model exactly.

Verification Check	Result
Original model test accuracy	0.9825
Pickle predictions match original	True
Joblib predictions match original	True

Difference Between Pickle and Joblib

- Pickle is Python's built-in general-purpose object serialization module. It works well for ordinary Python objects and smaller models.
- Joblib is commonly preferred for scikit-learn models because it is optimized for objects containing large NumPy arrays.
- Pickle is convenient when no extra dependency is desired; Joblib is often better for larger ML pipelines and numeric arrays.
- Both Pickle and Joblib files must only be loaded from trusted sources because loading serialized objects can execute unsafe code.

Part A/B Model Persistence Verification

Check	Result
Original accuracy	0.9825
Pickle predictions match	True
Joblib predictions match	True
Saved formats	.pkl and .joblib

Screenshot 1: Model persistence verification table.

Part C - End-to-End ML Pipeline

The mini-project uses the public UCI Wine Recognition dataset available through scikit-learn. This dataset is different from the Breast Cancer dataset used in Part A/B and contains chemical analysis measurements for three wine cultivars.

Dataset Property	Value
Dataset	UCI Wine Recognition dataset via scikit-learn
Rows	178
Columns after feature engineering	16
Missing values	0
Class counts	class_1: 71, class_0: 59, class_2: 48

Exploratory Data Analysis

- Class distribution was checked to understand the target balance across the three cultivars.
- A correlation heatmap was created to identify relationships among chemical features.
- Boxplots were used to compare selected feature ranges by wine class.
- A PCA scatter plot was used to visualize whether the classes form separable clusters in two dimensions.

Preprocessing

The original Wine dataset is fully numeric and has no missing values. To demonstrate a complete preprocessing pipeline, an additional categorical feature named alcohol_level was engineered from the alcohol column using quantile bins. The final pipeline still includes imputers so it remains robust if similar future data contains missing values.

Feature Type	Preprocessing Steps
Numeric features	Median imputation followed by StandardScaler
Categorical feature	Most-frequent imputation followed by OneHotEncoder
Train/test split	75% training and 25% testing with stratification

Model Training and Evaluation

A RandomForestClassifier was selected because it performs well on tabular classification tasks, captures nonlinear relationships, and does not require strong linear assumptions.

Metric	Value
Accuracy	1.0000
Macro F1-score	1.0000
Final saved model	outputs/models/part_c_wine_random_forest_pipeline.joblib

Classification Report and Interpretation

The following table summarizes class-wise precision, recall, F1-score, and support for the Wine mini-project test set.

Class	Precision	Recall	F1-score	Support
class_0	1.00	1.00	1.00	15
class_1	1.00	1.00	1.00	18
class_2	1.00	1.00	1.00	12
macro avg	1.00	1.00	1.00	45
weighted avg	1.00	1.00	1.00	45

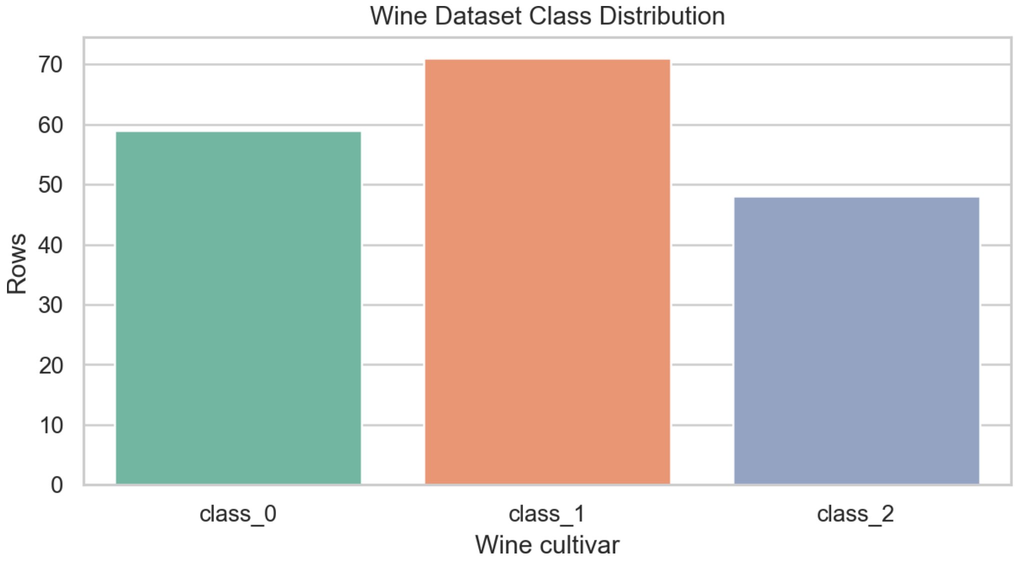
Result Interpretation

- Precision of 1.00 means all predicted samples for each class were correct in this test split.
- Recall of 1.00 means the model correctly identified all actual samples from each class.
- Macro F1-score of 1.00 shows balanced performance across the three wine classes.
- The strong results are consistent with the PCA plot and feature boxplots, which show clear class separation.

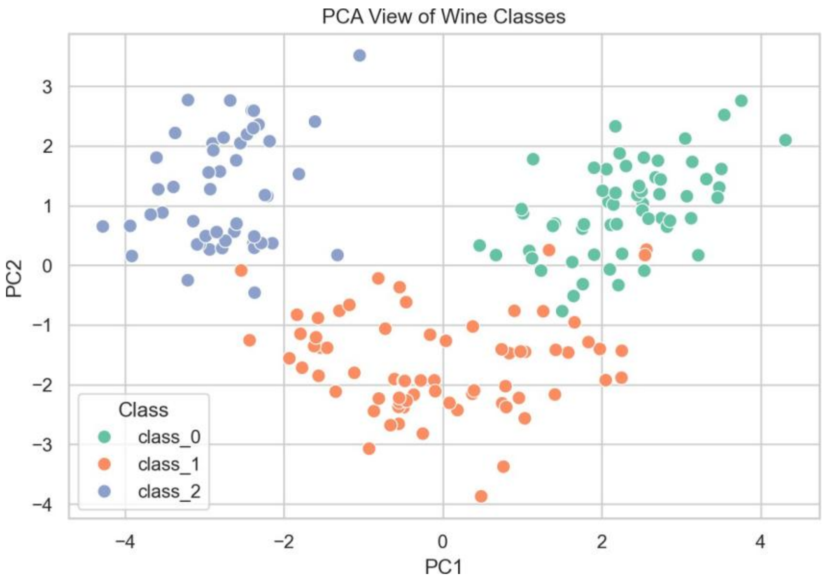
Saved Mini-Project Model

The final model was saved as a complete Joblib pipeline at outputs/models/part_c_wine_random_forest_pipeline.joblib. This file includes preprocessing and the trained RandomForestClassifier together.

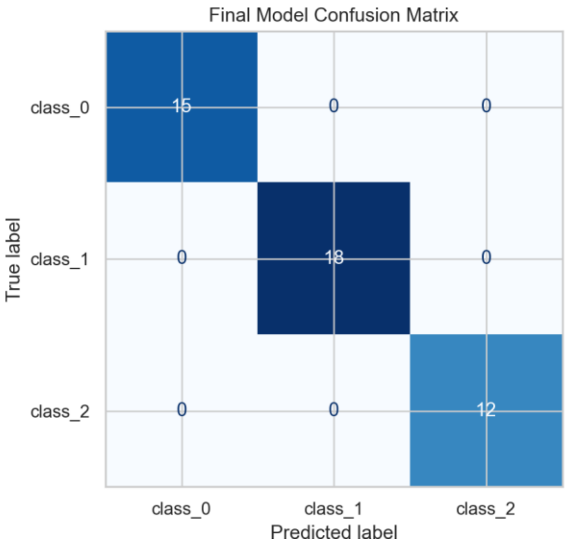
EDA and Evaluation Screenshots



Screenshot 2: Wine class distribution.

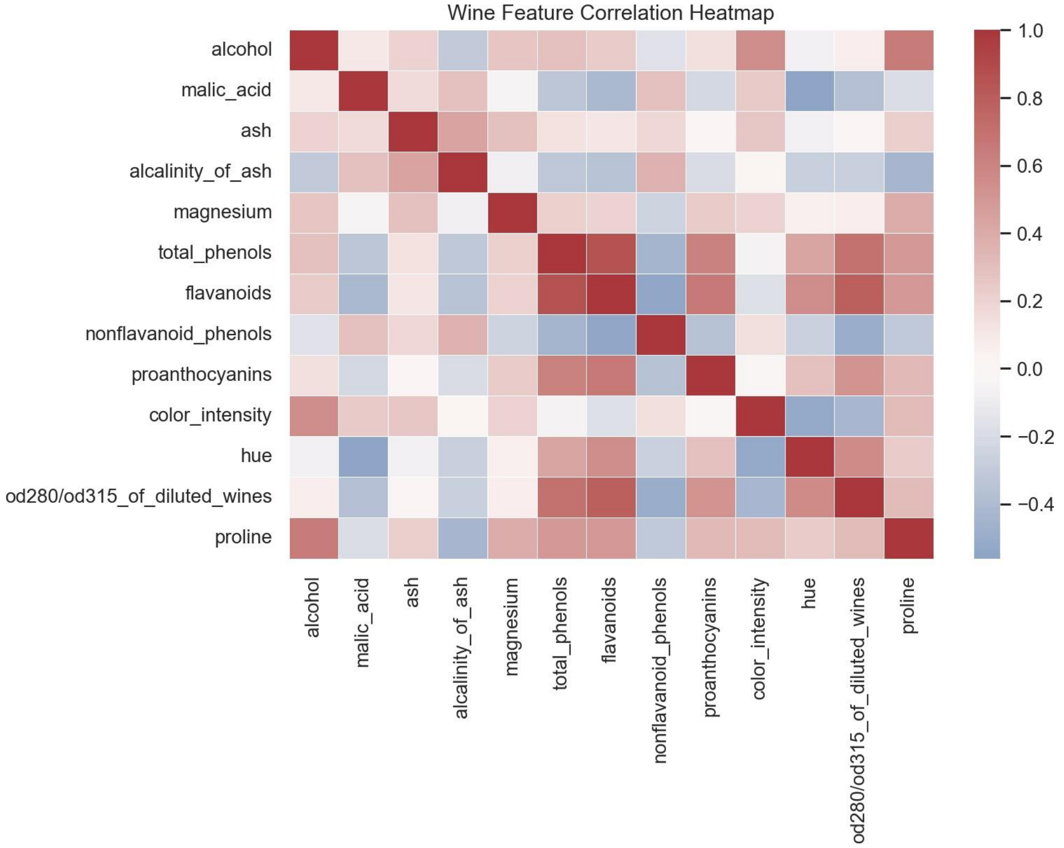


Screenshot 3: PCA plot showing class separation.

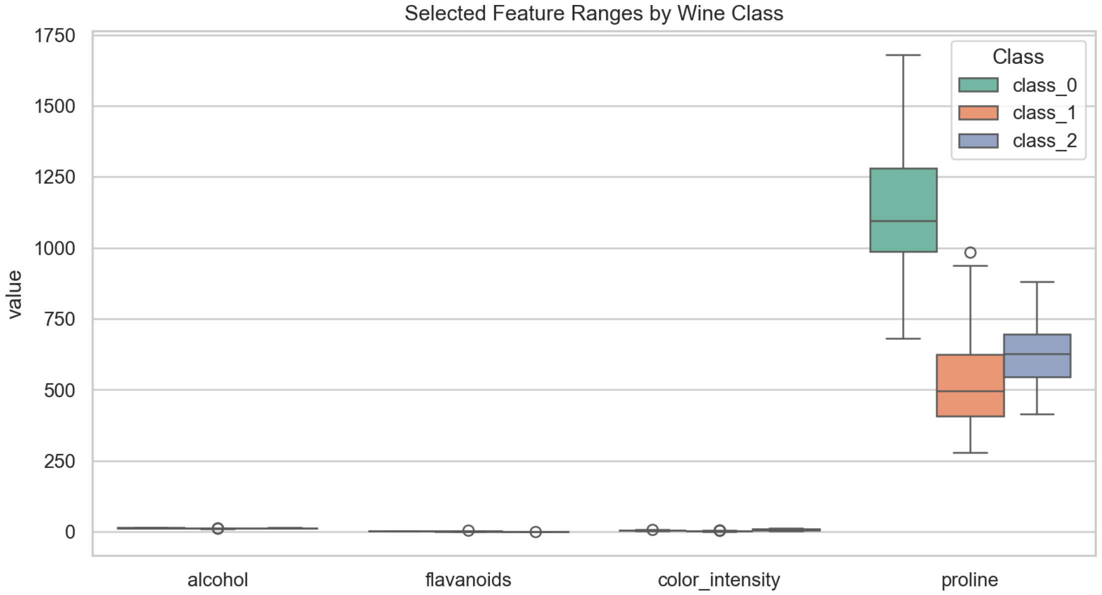


Screenshot 4: Final model confusion matrix.

Additional EDA Screenshots



Screenshot 5: Correlation heatmap for numeric wine features.



Screenshot 6: Selected feature ranges by wine class.

Part D - Model Saving & Documentation (15 Marks)

The final mini-project model was saved using Joblib as a complete scikit-learn Pipeline. Saving the entire pipeline is important because it preserves preprocessing steps and the trained classifier together, allowing the model to be reused consistently.

Key Findings

- The Wine dataset is clean and contains no missing values.
- Chemical measurements such as flavanoids, color intensity, proline, and alcohol help separate the three classes.
- The PCA visualization shows strong visual separation among classes.
- The final Random Forest pipeline achieved perfect performance on the selected stratified test split.

Challenges Faced

- The dataset is small, so a stratified train/test split was used to preserve target balance.
- The dataset is naturally numeric, so a simple categorical feature was engineered to demonstrate encoding.
- The pipeline was designed with imputers even though the dataset has no missing values, making the workflow more practical for real-world data.

Conclusion

This task demonstrates complete model persistence using Pickle and Joblib, verifies that loaded models reproduce the original predictions, and applies a full ML workflow from EDA through final model saving. The final deliverables include a notebook, saved model files, screenshots, and this formatted report.